

This code was first proposed in Ref. [18] using the polynomials

$$\begin{aligned} f(x, y) &= x^{25} + x^{26} + y^3, \\ g(x, y) &= y + y^2 + x^9, \end{aligned} \quad (69)$$

and implemented on an untwisted 30×6 torus. The stabilizers have a range of 9 in the x -direction (since x^{25} and x^{26} can be equivalently treated as x^{-5} and x^{-4}).

In contrast, our [[360, 12, 24]] code is simply the (3, 3)-BB code (Example 3), specified by the following polynomials:

$$\begin{aligned} f(x, y) &= x + x^2 + y^3, \\ g(x, y) &= y + y^2 + x^3, \end{aligned} \quad (70)$$

placed on a twisted 6×30 torus with lattice vectors $\vec{a}_1 = (0, 30)$ and $\vec{a}_2 = (6, 6)$. For physical realization, once we have the architecture for the stabilizers of the (3, 3)-BB code, we can generate optimal generalized toric codes on various lattices, as listed in Eq. (59).

Similarly, the physical construction of the (3, -3)-BB code (Example 4) can generate the quantum LDPC codes listed in Eq. (58). By comparing the stabilizers in Tables I, II, III, and IV with those in the literature, we observe that twisted tori generally reduce the range of stabilizers, making experimental realization more feasible.

C. Relation to one-dimensional generalized bicycle codes

We present another example from Table III, the [[254, 14, 16]] code, which achieves $kd^2/n = 14.11$. This code is defined on a twisted 1×127 torus with lattice vectors $\vec{a}_1 = (0, 127)$ and $\vec{a}_2 = (1, 25)$. The associated polynomials are:

$$\begin{aligned} f(x, y) &= 1 + x + x^{-1}y^{-3}, \\ g(x, y) &= 1 + y + y^{-6}. \end{aligned} \quad (71)$$

The code is local on the twisted torus, as the range of each stabilizer is small relative to the total system size n . Since the twisted torus is narrow in the x -direction, we can remove the x -direction periodicity by using the polynomial $xy^{25} - 1$ to cancel the x -dependence. Therefore, we can reduce the code to a non-local one-dimensional quantum code. This transformation yields the following polynomials:

$$\begin{aligned} f(y) &= 1 + y^{22} + y^{102}, \\ g(y) &= 1 + y + y^{121}, \end{aligned} \quad (72)$$

with a periodic boundary condition $y^{127} - 1 = 0$. In these one-dimensional codes, the Gröbner basis in Theorem 3 reduces to the gcd (greatest common divisor) for univariate polynomials, simplifying to the following expression:

$$\begin{aligned} k &= 2 \dim \left(\frac{\mathbb{Z}_2[y, y^{-1}]}{\langle f(y), g(y), y^l - 1 \rangle} \right) \\ &= 2 \deg (\gcd (f(y), g(y), y^l - 1)), \end{aligned} \quad (73)$$

which precisely matches Proposition 1 in Ref. [124], which computes the logical dimension of the generalized bicycle (GB) codes [125]. For each value of n , we can apply the same procedure to the generalized toric codes on the twisted $1 \times \frac{n}{2}$ tori:

$$\vec{a}_1 = (0, \frac{n}{2}), \quad \vec{a}_2 = (1, \gamma), \quad \text{with } 0 \leq \gamma < \frac{n}{2}, \quad (74)$$

to induce the corresponding one-dimensional generalized bicycle codes. The results are summarized in Table V, VI, and VII in Appendix A.

For comparison, consider the GB code described in Ref. [124]. The polynomials for this GB code are:

$$\begin{aligned} f(y) &= 1 + y^{15} + y^{20} + y^{28} + y^{66}, \\ g(y) &= 1 + y^{58} + y^{59} + y^{100} + y^{121}, \end{aligned} \quad (75)$$

defined on a cycle of length $l = 127$. This GB code uses weight-10 stabilizers to achieve better code parameters [[254, 28, $14 \leq d \leq 20$]].

IV. DISCUSSION AND FUTURE DIRECTIONS

We have introduced a topological order perspective to studying quantum error-correcting codes on tori. From the algebraic structure of anyons, the logical dimension k can be determined by counting independent anyon types. We showed that this corresponds to the dimension of the quotient ring R/I , where the ideal $I = \langle f(x, y), g(x, y) \rangle$ is generated by the stabilizers. This provides a systematic approach to characterizing the code space. Our framework naturally incorporates (twisted) periodic boundary conditions, enabling the construction and characterization of new quantum LDPC codes. To ensure computational feasibility, we employed Gröbner basis techniques, enabling a systematic analysis of generalized toric codes up to $n \leq 400$ physical qubits. The versatility of our method is reflected in the discovery of novel qLDPC codes listed in Tables I, II, III, and IV. These results illustrate the power of a ring-theoretic approach in advancing the understanding of topological quantum codes, paving the way for future explorations in both theory and practical implementation.

Future work could extend this investigation to larger system sizes (higher n), as these may yield improved codes. Given that our search algorithm is fully parallelizable, supercomputers or computer clusters could be employed to examine all generalized toric codes within $n \leq 500$ or higher—scales that are comparable to the number of physical qubits in state-of-the-art experimental platforms [126–131]. The primary bottleneck, however, is the computation of code distances. When n reaches a few hundred and d exceeds 20, the probabilistic algorithm for computing the code distance may not be reliable and could only yield an upper bound for d .

Alternatively, one could explore different forms of